A Comprehensive Review of DRAM NBTI Issues and Solutions

Faxian Shan[1*], Pinghui Li[1], Ming Wang[1], Wei Tang[1], Yue Cao[1], Shun Wang[1], Hu Cai[1], Chuangming Hou[1], and Zhiyong Jiang[1]
[1]ChangXin Memory Technologies, Inc., Hefei 230088, China
*Corresponding author: shanfaxian@163.com; Tel / Fax: +86-18986180017

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***Abstract*—A comprehensive review of NBTI issues and their preventions were summarized over the past years. The NBTI related process issues, degradation mechanism, and process solution were systemically summarized. We also analyzed some NBTI risk circuits and provided improve suggestions to enlarge DRAM circuit NBTI tolerance. To disclosure the circuit level real risk, a new methodology was investigated with quiz-static DC stress on ring oscillator. Its degradation exhibits a good co-relation to that of single device level NBTI performance, and can be used to guide the definition of NBTI specification. This work provides some valuable insights to DRAM NBTI solutions.**

Keywords—DRAM, Invertor, NBTI Mechanism, pMOSFET, Process Change, Ring Oscillator

I. Introduction

Gate dielectric thickness thinning down to boost device performance has caused a high electric field and posed a great reliability concern on pMOSFET negative bias temperature instability (NBTI) related reliability degradation [1,2]. Dynamic random access memory (DRAM) circuits are susceptive to this degradation, and there have been several cases reported on NBTI related failure [3,4]. To improve the retention characteristic and overall device performance, hydrogen annealing is widely used on memory processing. However, the over incorporated hydrogen would further enhance NBTI degradation [1,5]. It exhibits a strong process dependence, and there are still some controversy on the underlying mechanism. Besides, we lack of a systematical review about this effect on DRAM circuits. This work summarized some DRAM NBTI issues and process reasons, disclosure the mechanism and solutions of related product reliability failure, and highlighted the susceptive periphery circuits. It also demonstrated a circuit level NBTI assessment methodology with "quasi-static" stress on related ring oscillator.

II. Experimental Details

We characterized pMOSFET with a poly gate and a SiON dielectric about 20 nm thickness, which are fabricated with commercial 19 nm DRAM technology process. The device size is 3 μm in width and 0.05 μm in channel length. After the formation of basic CMOS structure, unique processes of DRAM storage capacitor have introduced several adversary effects to the devices. Then the back end contact and metal layers in commercial 55 nm logic technology node were patterned to achieve the electrical connection function. The final alloy at about 400 ºC was implemented after the deposition of hydrogen-rich passivation layers, which introduced a huge amount of hydrogen to improve retention performance and partially mitigated the device reliability risk. Single device level NBTI stress was applied with the gate electrode held at a low constant negative bias (ranging from -1.6 V to -2.0 V) under the ambient temperature of 100 ºC, while the source/drain and n-well electrode were all grounded. The measurement was conducted on a Keysight B1500A semiconductor device analyzer with a temperature well controlled prober station to ±1 ºC. To analyze the circuit level NBTI specification, a new characterization methodology was also introduced on a ring oscillator circuit featuring of 21 stages of invertor and 4 stages of readout buffer. The "quasi-static" NBTI stress was applied to Vdd pin when turned off the ring oscillator enable pin by using the above tool, while the readout was achieved when turned on the enable pin and readout the waveform by using a Keysight MXR608A mixed signal oscilloscope. To setup the correlation to device level test, circuit level NBTI experiment was also conducted at 100 ºC with the same wafer.

III. Results And Discussion

A. Process induced NBTI issues

Fig.1 shows that the voltage dependence of device level NBTI lifetime well follows the classic power law model even at different processing vintage. The red point represents the normal use voltage condition which is about 1.1 V. It is obvious that "Pre Pad Out Alloy" wafer exhibits a much smaller lifetime (four orders less) than that of "Post Pad Out Alloy" wafer. Besides, a smaller voltage dependence slope also implies that there are other mechanism dominating NBTI electrochemical reaction in "Pre Pad Out Alloy" wafer. To analyze the underlying mechanism, a detailed comparison of these two alloy manner [6] was summarized on Fig. 2. "Pre Pad Out Alloy" alloyed with the nitride passivation layer fully covering the wafer before pad open step. As well known, back end metal related deposition and etching loops can incorporate many water and chemicals, which are hardly to be fully removed due to the very shorten processing time of each steps. In "Pre Pad Out Alloy" wafer, the moisture from these layers cannot diffuse out because the nitride passivation is nearly impermeable to moisture. Instead, much of it will diffuse to the oxide/silicon interface along with the hydrogen, and enhance the NBTI electrochemical reaction rate due to the increased number of water related defect [7]. Although it induced a serious NBTI lifetime degradation and speed related timing loss, this case is irrecusable because it provides an efficient solution to improve DRAM access transistor subthreshold swing and product retention performance due to the huge amount of hydrogen incorporation from nitride passivation layer [8]. While "Post Pad Out Alloy" etch pad window firstly, the moisture and hydrogen can easily diffuse to ambient through the opened pad window when under the following alloy loop. Owing to the limited hydrogen compensation effect, access transistor and product retention performance with this alloy manner can hardly meet the competitive marketing requirements. However, this case exhibits a much better NBTI reliability and speed timing

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performance due to less water related defect at the oxide/silicon interface.

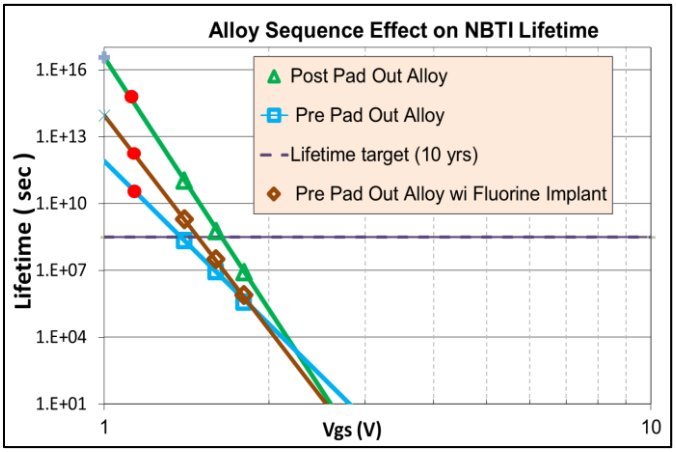

Fig. 1. Alloy sequence effect on NBTI lifetime and its voltage dependence.

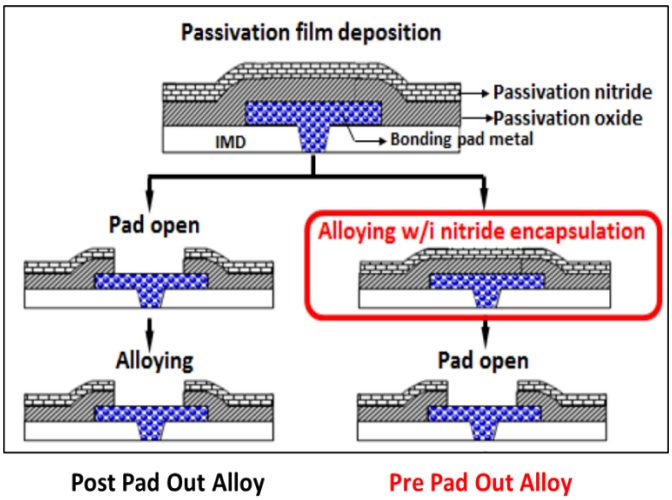

Fig. 2. Process catron of alloy and padout process.

To mitigate the speed timing loss due to "Pre Pad Out Alloy" NBTI degradation, an innovative solution was developed to add additional fluorine during poly gate implantation. After the following thermal anneal process, fluorine ion can reach to oxide/silicon interface and form the stable Si-F bonding which possesses a much larger rupture energy than Si-H bonding. Besides, fluorine ion can react with Si-O bonding, then release O ion to furtherly oxidize the oxide/silicon interface, and finally enlarge the oxide thickness [2]. The stable Si-F bonding and the enlarged oxide thickness can greatly reduce NBTI electrochemical reaction rate and enlarge reliability lifetime to an acceptable level, as shown on Fig. 1. Therefore, a good balance between DRAM retention performance and the speed timing loss due to NBTI degradation has been achieved with this novel solution.

Pad out etching loop usually makes a top contribution to gate oxide plasma damage issue with the largest plasma energy in normal DRAM process. The related process steps were listed in Fig. 3a. In this paper, we only focus on the asher step under a plasma environment which is used to clean the organic photoresistor residue and polymer byproduct post etching process. Without passivation protection, plasma charge can directly touch the metal layer and accumulate at the gate electrode. Although pMOSFET usually has a protection diode with p+ implant in n type well to protect the potential damage effect, this diode is reversed when posed to a large amount of negative plasma charging during the ahser steps, and cannot efficiently discharge the accumulated charging even at a high temperature conidition, as illustrated on Fig. 3b. When the load is up to a certain level, the large voltage difference can initiate the Fowler-Nordheim tunneling [9] current across the gate oxide, and results in related oxide bulk defects and interface traps. Even interface traps can be temporarily repaired during alloy process, they will degrade easily when under a reliability stress condition.

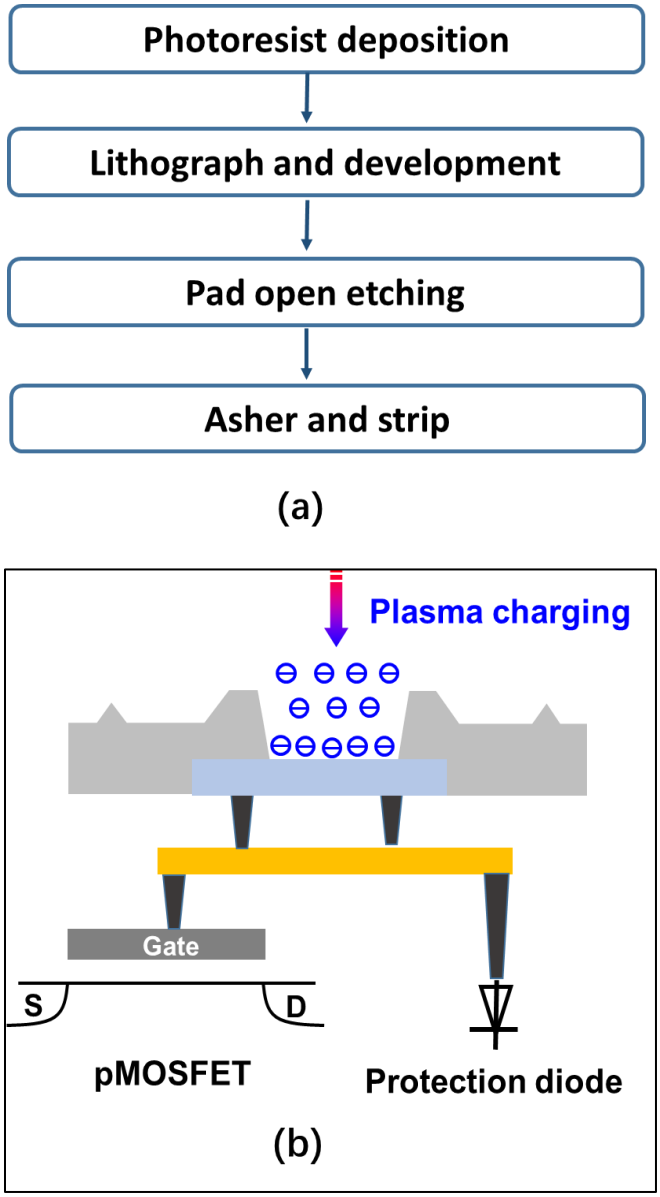

Fig. 3. (a) Pad out etching loop related process steps. (b) pMOSFET plasma damage mechanism during asher process post pad out etching.

When the reactant gas of asher process is changed from a nitrogen & oxygen mixture to a nitrogen & hydrogen mixture, the polymer cleaning effect has been greatly improved. However, this change results in a severe NBTI degradation with a larger Idsat degradation value (8.9% vs. 5.1% at 1000s) and a larger time slope (0.21 vs. 0.1) in Fig. 4. The additional ionic hydrogen incorporation and plasma potential change due to reactant gas change should account for this issue, since there is more interface weak point precursors was formed.

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Therefore, process plasma should be well controlled to increase NBTI immunity of high quality DRAM products.

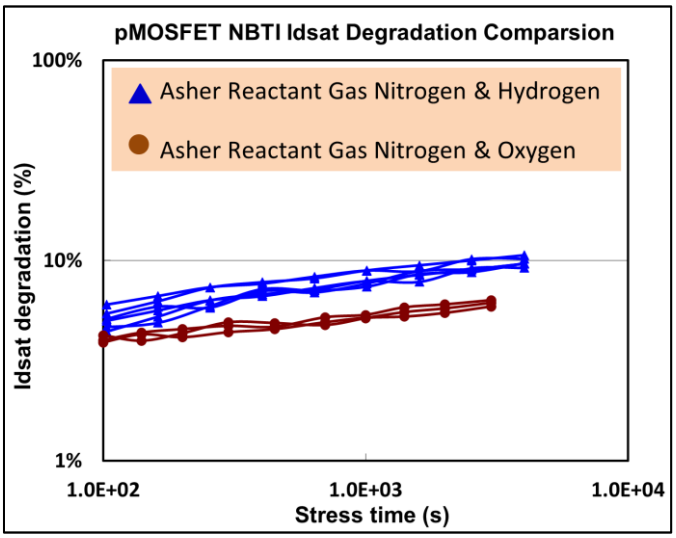

Fig. 4. Gas change of passviation etching asher loop enahanced NBTI degradation

B. *Circuct level NBTI issues*

DLL (delay locked loop) circuit is used to provide a stable delay time between DQ (data queue) and DQS (data queue strobe) signal to guarantee data reading integrity in the modern DDR SDRAM (Double data rate synchronous dynamic random access memory) chip. This circuit usually has hundreds stage of CMOS invertor in serials ranging from S1 to Sn+1, as shown in Fig. 5a. We have once encountered a serious reading data eye window shift issue, one of DRAM most important parameter, due to this circuit pMOSFET NBTI degradation under product level HTOL (high temperature operation lifetime) test at 125 ºC and Vdd = 1.3V. Before this test, even and odd data eye window are in the same level. When after stress, the odd data eye degrades heavily, and the large imbalance about 150 ps between the even and odd eye can be seen from the blue boxes in Fig. 6. After systematic analysis through risky device verification and circuit simulation, an underlying degradation model in Fig. 5b was disclosed to explain this imbalance. The above experiment was conducted at DLL circuit off stress mode. This circuit keeps at a DC bias condition with only the odd invertors subjected to NBTI stress (at low gate bias). Although the even invertors are at pMOSFET PBTI stress condition (at high gate bias), this effect is negligibly small when compared to NBTI degradation at SiON gate dielectric technology [10]. The rising edge keeps suffering timing delay when passing through each odd invertor due to the above device degradation, while the failing edge keeps no changed. Therefore, a serious duty cycle shift from 50%/50% to 35%/65% can be detected from the waveform. However, this circuit always keeps AC toggle and not suffer above DC condition when DRAM normal application condition. Then, the odd device NBTI degradation can be greatly reduced at customer end due to the well-known AC recovery effect [11] of NBTI electrochemical reaction. Therefore, we modified HTOL stress pattern with DLL circuit turned on by AC toggle signal referring to the real usage case. The even odd imbalance has been greatly reduced to less than 50 ps (see the green boxes in Fig. 6).

Another NBTI susceptive circuit is the equalizer driver, which also consists of many invertor trees in serials. The equalizer circuit plays an important role, which is used to precharge bit line voltage to a certain reference level before data writing operation. The device performance in equalizer driver path determines bit line precharge efficiency, characterized as tRP timing (precharge command period). If the driver device has degraded heavily, tRP timing would be enlarged to an unacceptable level to disturb next loop data writing accuracy. Furthermore, the equalizer voltage should be 0.3 ~ 0.5 V higher than normal Vdd to pass through the bit line voltage successfully, indicating that these devices suffer more serious NBTI stress than normal core transistors due to a higher electric field. Considering DRAM application characteristics, the most of equalizer driver devices are under standby mode. Therefore, even or odd pMOSFET in this circuit continuously suffer DC NBTI bias stress among the most of life cycle at customer end. Although they can change the bias status for a very short period when receive a precharge command, they returned to DC NBTI condition again when this command finished. Therefore, equalizer driver pMOSFET sustains the most serious NBTI stress in DRAM chip. There was once a tRP timing degradation issue at burn in stress, due to the above degradation model. It was solved by a new type of device featuring with a high NBTI immunity, which will be published elsewhere. With the gate dielectric continuously thinning down, the breaking through innovation is needed for this circuit design to ensure enough reliability margin.

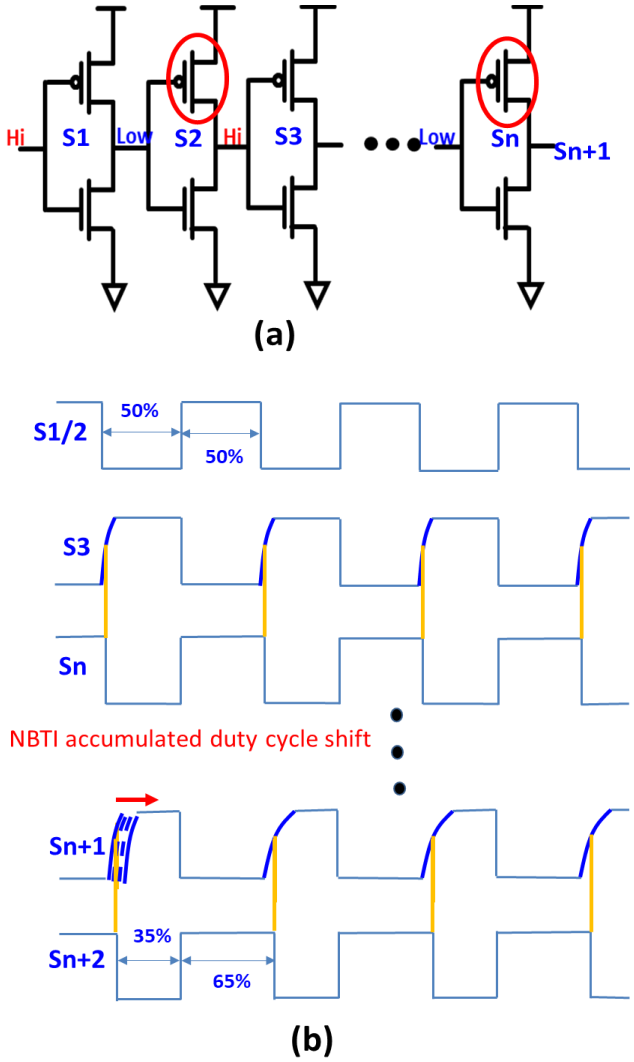

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Fig. 5. (a) DLL circuit consists of hundreds of invertors in serials. (b) imbalant NBTI stress cause duty cycle shift.

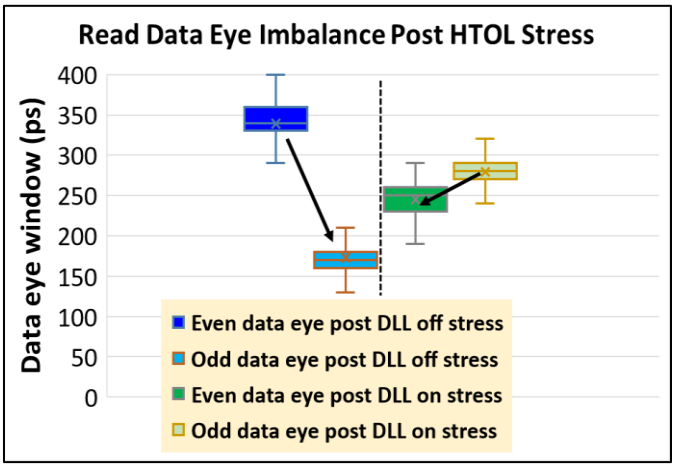

Fig. 6. Read data eye window post HTOL stress at DLL off and DLL on mode

C. Circuit level NBTI characterization

It is quite challengeable to setup a direct correlation between single device level NBTI and DRAM product timing degradation. Ring oscillator (RO) with investor trees in serials has been already used to investigate device hot carrier degradation effect on circuit performance for many years [12,13], and to monitor CMOS circuit speed performance. Its schematic is very similar to that of above mentioned DLL and equalizer driver circuits. As far as we known, no studies have ever been published to simulate pMOSFET NBTI degradation by it, because NBTI can quickly get recovered during oscillation test. In this paper, we developed a "quasi-static" NBTI stress method with it. As illustrated in Fig. 7, ring oscillator NBTI stress voltage 1.5/1.8/2.0V was applied to Vdd pin with the oscillation enable pin turned off. Therefore, the AC recovery effect was eliminated during the stress period. The interval readout were achieved by turning on the enable pin and recording the waveform parameters from an oscilloscope. Under high temperature and high Vdd "quasi-static" stress mode in this paper, pMOSFET NBTI should have dominating contribution to ring oscillator degradation than other device degradation mechanisms. Therefore, a strong correlation was expected between single device and ring oscillator NBTI. Fig. 8 confirms the correlation between pMOSFET Idsat and related ring oscillator period delay time (tPD) degradation. Their degradation increase with the stress voltage and stress time, and the dependence are summarized as Equation (1) and (2), respectively. Ring oscillator tPD exhibits a larger voltage and time dependence factor due to interval readout delay and recovery effect. Ring oscillator readout delay time is much longer (in second level vs. in millisecond level) than that of pMOSFET NBTI test, and it suffers AC recovery effect during waveform read oscillation. With the above correlation, we can still easily define single pMOSFET NBTI reliability specification based on ring oscillator circuit speed boundary. If design tolerance on ring oscillator tPD degradation is 5% at normal Vdd use condition, the target lifetime can be calculated from Equation (2). After incorporate this lifetime and Vdd value to Equation (1), we can extrapolate pMOSFET Idsat degradation value as the desired specification, which is usually tighten than the foundry widely used value (10%). Combining this specification and device level test convenience and high efficiency, we can qualify and monitor process health in a more realistic and efficient way. Further endeavors are still under progress to minimize the ring oscillator readout delay time.

$$\Delta\ \mathrm{Idsat} = \frac{Vstress^{5.49} \times t^{0.26}}{e^{9.34}} \quad (1)$$

$$\Delta\ \mathrm{tPD} = \frac{Vstress^{5.83} \times t^{0.34}}{e^{10.65}} \quad (2)$$

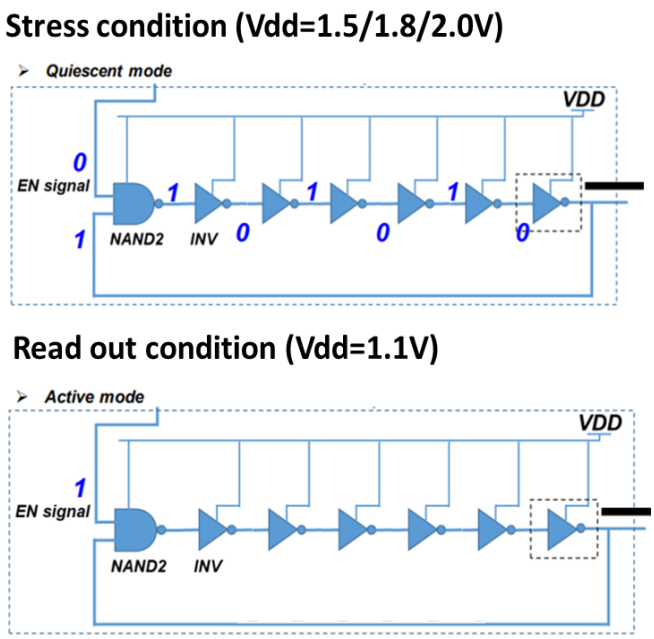

Fig. 7. Ring oscillator NBTI stress conidition and readout conidition.

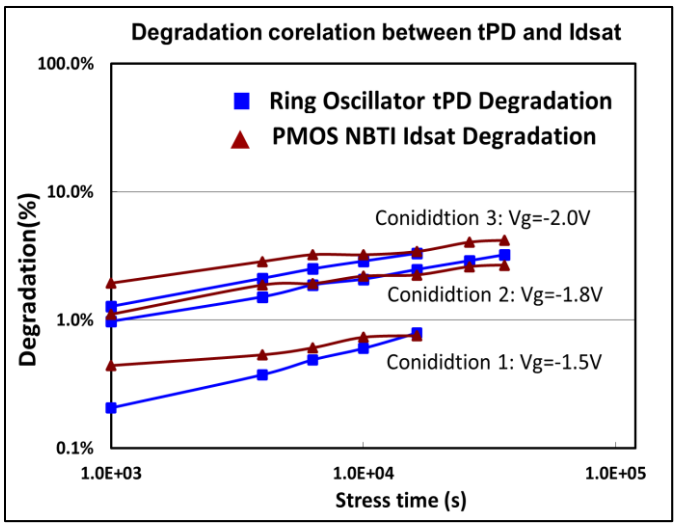

Fig. 8. A strong correlation exists between pMOSFET Idsat and related ring oscillator tPD degradation.

IV. Conclusions

In summary, the single device and circuit level NBTI issues were summarized and analyzed in this paper. "Pre Pad Out Alloy" process degrade NBTI since moisture enhance electrochemical reaction as a catalyst. Fluorine implantation from poly gate can fix this issue by forming stable Si-F bond at interface and enlarging gate dielectric thickness. Besides, pad out etch loop plasma with high energy plays an important role to determine interface damage and NBTI degradation.

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HTOL reading data eye shift issue caused by DLL circuit NBTI imbalance was solved by AC toggle signal. Equalizer driver is also an susceptive circuit with a narrowest NBTI margin, therefore it needs the breaking through innovation from both design and process sides. To corelate single device NBTI degradation to circuit speed degradation, we demonstrate a circuit level stress method and use it to define single device NBTI reliability specification. Then, we can qualify and monitor process in a more realistic and efficient way.

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